TCS NQT 2027 · SECTION 2 OF 5 · PART A · FOUNDATION

Reasoning Ability Complete Guide

Every topic explained from zero: rules, formulas, diagrams, solved examples and practice questions.

11	44+	110+	20
topics from basic to advanced	solved examples	practice questions	question mini test

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About this guide

Reasoning Ability is the second section of the TCS NQT Foundation test. It has 20 questions in 25 minutes (about 75 seconds each), moderate difficulty, no negative marking, and some questions are TITA (you type the answer).

How each topic is organised

- **Start from zero:** the idea in plain words.
- **Formulas and rules:** a yellow box to revise quickly.
- **Diagrams:** pictures, charts and tables.
- **Solved examples:** step by step.
- **Shortcuts and mistakes** to avoid.
- **Practice questions:** Basic, Medium and Advanced. Answers are at the end.

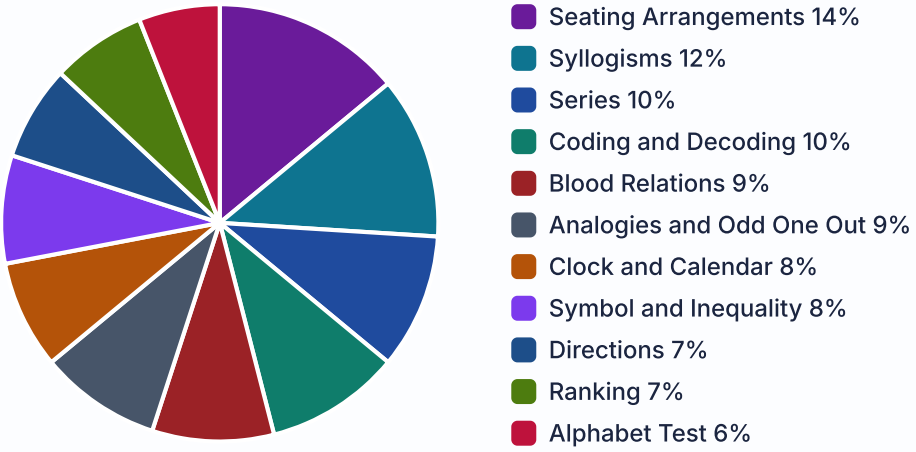
How to study

- Read one topic, then solve its practice set without looking at the answers.
- Check answers and note wrong ones in an error log.
- Revise the formula boxes daily.
- After all topics, take the mini test under the real time limit.
- Follow the priority: do **Must-do** topics first.

Topic priority and contents

1. Number and Alphabet Series	Must-do	2. Coding and Decoding	Must-do
3. Blood Relations	Must-do	4. Directions and Distances	Important
5. Seating Arrangements	Must-do	6. Syllogisms	Must-do
7. Clock and Calendar	Important	8. Ranking and Ordering	Important
9. Alphabet Test	Important	10. Symbol Operations and Inequalities	Important
11. Analogies and Odd One Out	Good to know		

Where to spend your practice time



Suggested share of practice time per topic. This is our preparation guidance, not an official TCS weightage.

Section strategy

Reasoning Ability has 20 questions in 25 minutes (about 75 seconds each), moderate difficulty and no negative marking.

Time plan

Phase	Time	What to do
Pass 1	14 min	Series, coding, relations, directions, clock and calendar
Pass 2	8 min	Puzzle sets (seating) and syllogisms
Final	3 min	Guess any left blank

Rules of this section

- Question switching allowed inside the section.
- No section switching after time ends.
- No negative marking: attempt everything.
- Use rough paper for diagrams and tables.

Speed habits

- Solve series and coding in under 40 seconds.
- For puzzles, draw the table before reading all clues twice.
- Use Venn diagrams for every syllogism.
- Keep a mini compass for direction questions.

Where marks are lost

- Left and right confusion in seating and directions.
- Treating 'Some' as 'All' in syllogisms.
- Starting puzzles without a diagram.
- Spending more than 2 minutes on one question.

01 REASONING ABILITY · MUST-DO
Number and Alphabet Series

A series follows a hidden rule. Your job is to discover the rule from the first few terms and apply it once more. This is the fastest scoring topic in Reasoning because the question is short and the rule is usually one of a handful of types.

Start from zero

The common patterns

Arithmetic: add or subtract the same number (3, 7, 11, 15 → +4). **Geometric:** multiply or divide by the same number (2, 6, 18, 54 → ×3). **Square or cube:** 1, 4, 9, 16 or 1, 8, 27, 64. **Square ± something:** 2, 5, 10, 17, 26 is $n^2 + 1$. **Increasing gaps:** 2, 3, 5, 8, 12 has gaps 1, 2, 3, 4. **Alternating:** two series woven together or +3, −2, +3, −2. **Primes and Fibonacci:** 2, 3, 5, 7, 11 and 1, 1, 2, 3, 5, 8.

How to find the rule fast

First write the gaps. If the gaps are equal it is arithmetic. If the gaps themselves form a pattern (1, 2, 3 or 2, 4, 6) use a second row of gaps. If the numbers roughly double each time, test multiplication. If numbers look like squares or cubes, compare with n^2 and n^3. If it still fails, split into odd-position and even-position terms.

Difference method: write the gap between neighbours



Gaps 1, 2, 3, 4, 5 … grow by 1 each time → next gap is 6 → 23 + 6 = 29

Other patterns: ×2, ×3 (geometric) · squares 1, 4, 9 · cubes 1, 8, 27 · alternating (+3, −2, +3 …) · primes 2, 3, 5, 7 · Fil

Series: find the gaps or the multiplier

Letter series

Convert letters to positions (A=1 … Z=26), find the number pattern, then convert back. B, E, H, K has positions 2, 5, 8, 11 (+3), so the next is 14 = N. Remember opposite letters add to 27 and wrap around after Z when counting forward.

Missing and wrong term

In a missing-term question, find the rule from the visible terms. In a wrong-term question, find the rule from the majority of terms and spot the one that breaks it.

Formula and rule box

- Arithmetic: next = last + d
- Geometric: next = last × r
- Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100
- Cubes: 1, 8, 27, 64, 125, 216
- Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
- Fibonacci: each term = sum of previous two
- Letter position: A=1 … Z=26; opposite = 27 − position

Step-by-step method

1. Write the gaps between consecutive terms.
2. If gaps are not constant, write the gaps of the gaps or the ratios.
3. Check squares, cubes and primes.
4. Try odd and even positions separately.
5. Apply the rule once and verify with the given terms.

Solved examples

Example 1. Find the next term: 3, 7, 11, 15, ?

Solution: Gaps are 4, 4, 4. Next = 15 + 4 = 19.

Answer: 19

Example 2. Find the next term: 2, 3, 5, 8, 12, 17, ?

Solution: Gaps: 1, 2, 3, 4, 5. Next gap is 6, so 17 + 6 = 23.

Answer: 23

Example 3. Find the next letter: B, E, H, K, ?

Solution: Positions 2, 5, 8, 11 (+3). Next = 14 = N.

Answer: N

Example 4. Find the next term: 2, 6, 18, 54, ?

Solution: Each term is ×3. Next = 54 × 3 = 162.

Answer: 162

Shortcuts

- Learn squares and cubes so you spot them at once.
- If the numbers rise slowly, think addition; if they rise quickly, think multiplication or powers.
- Two interleaved series: check terms 1, 3, 5 and 2, 4, 6 separately.

Common mistakes

- Stopping after checking only one gap.
- Forgetting that letters wrap around after Z.
- Missing an alternating pattern like +3, −2.

Practice questions: Number and Alphabet Series

1. 5, 9, 13, 17, ? Basic
2. 3, 6, 12, 24, ? Basic
3. 1, 8, 27, 64, ? Basic
4. 2, 5, 10, 17, 26, ? Basic
5. 4, 6, 9, 13, 18, ? Medium
6. 2, 3, 5, 7, 11, 13, ? Medium
7. A, C, F, J, ? Medium

8. 1, 1, 2, 3, 5, 8, ? Medium

9. 2, 12, 36, 80, 150, ? Advanced

10. 7, 10, 8, 11, 9, 12, ? Advanced

Answers: see the answer key at the end of this guide.

02 REASONING ABILITY · MUST-DO
Coding and Decoding

Coding questions hide a rule that turns words into other words or numbers. Once you see whether the rule is a shift, a reversal or a substitution, the rest is mechanical.

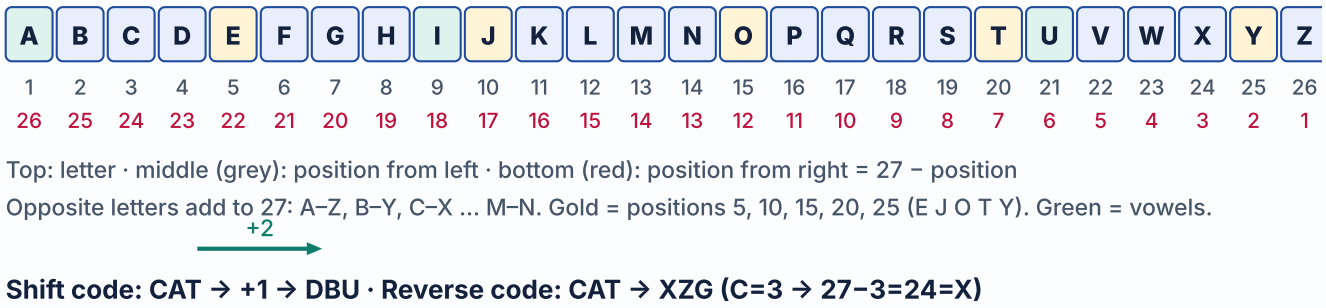
Start from zero

Letter shifting

Each letter moves forward or backward by a fixed number. CAT with +1 becomes DBU. Check the first two letters to find the shift, then apply it to the rest. A second pattern is an alternating shift (+1, −1, +1 ...).

Reverse alphabet (opposite letters)

Replace each letter by its opposite: A↔Z, B↔Y, C↔X ... The positions add to 27. So CAT becomes XZG: C is 3 and 27 − 3 = 24 = X; A → Z; T is 20 and 27 − 20 = 7 = G.



Top: letter · middle (grey): position from left · bottom (red): position from right = 27 − position

Opposite letters add to 27: A–Z, B–Y, C–X ... M–N. Gold = positions 5, 10, 15, 20, 25 (E J O T Y). Green = vowels.

The alphabet with positions from left and right

Position values

Convert letters to numbers using A = 1 ... Z = 26. TABLE becomes 20, 1, 2, 12, 5. A sum or product of positions can also be the code.

Word and number substitution

Compare two coded sentences with one common word to find the code of the shared word. Example: 'red blue' = 'ab cd' and 'blue green' = 'cd ef'. The common word blue = cd, so red = ab and green = ef.

Matching from given codes

If ROSE = 6821, CHAIR = 73456 and PREACH = 961473, read off each letter's code (R=6, O=8, S=2, E=1, C=7, H=3, A=4, I=5, P=9) and assemble any new word from these letters.

Formula and rule box

- Opposite letter = 27 − position
- Forward shift: new position = old + k (wrap after 26)
- Alternating shift: +k, −k, +k, ...
- Position value: A=1, B=2, ... Z=26
- Common-word method for sentence codes
- Check the first two letters to detect the rule

Step-by-step method

1. Write the original and coded word one under the other.
2. Compare letter by letter to see +, − or opposite.
3. Test the same rule on the other letters.
4. Apply it to the target word and check the answer is sensible.

Solved examples

Example 1. If MANGO is coded NBOHP, how is APPLE coded?

Solution: Each letter is +1 (M→N, A→B, N→O, G→H, O→P). APPLE → BQQMF.

Answer: BQQMF

Example 2. Write CAT in the reverse-alphabet code.

Solution: C(3) → 24 = X, A(1) → 26 = Z, T(20) → 7 = G.

Answer: XZG

Example 3. If ROSE = 6821, CHAIR = 73456 and PREACH = 961473, find the code of SEARCH.

Solution: S=2, E=1, A=4, R=6, C=7, H=3.

Answer: 214673

Example 4. In a code, 'red blue' = 'ab cd' and 'blue green' = 'cd ef'. What is the code of red?

Solution: Blue is common and equals cd. So red = ab.

Answer: ab

Shortcuts

- Opposite letters add to 27, so subtract the position from 27.
- Memorise EJOTY (5, 10, 15, 20, 25) as anchors.
- For sentence codes, find the common word first.

Common mistakes

- Forgetting to wrap around (Y + 2 = A, not 27).
- Applying the rule to only part of the word.
- Mixing forward and backward shifts.

Practice questions: Coding and Decoding

1. If BOOK = CPPL, find the code of LAMP. Basic
2. Find the reverse-alphabet code of DOG. Basic
3. Find the sum of positions of the word BAD (A=1). Basic
4. If shift is +2, code DOG. Basic
5. If shift is −1, code FIRE. Medium
6. TABLE is coded 20 1 2 12 5. Code CHAIR in the same way. Medium
7. Find the value of CAT if Z = 1, Y = 2 ... A = 26. Medium
8. Using +1, −1, +1, −1 alternately, code MOON. Medium

9. If +2 gives LOGIC = NQIKE, code BRAIN. Advanced

10. If ROSE = 6821, CHAIR = 73456, PREACH = 961473, find the code of SEARCH. Advanced

Answers: see the answer key at the end of this guide.

03 REASONING ABILITY · MUST-DO
Blood Relations

Blood-relation puzzles describe family links in words. Translate each sentence into a small family tree and the answer becomes a matter of reading the tree.

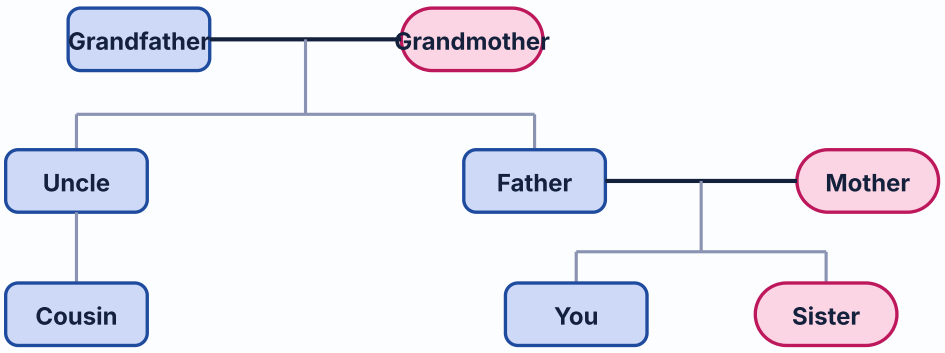
Start from zero

Standard relations

Father's or mother's father = grandfather. Father's or mother's mother = grandmother. Father's brother = paternal uncle. Mother's brother = maternal uncle. Father's or mother's sister = aunt. Brother's or sister's child = nephew (boy) or niece (girl). Uncle's or aunt's child = cousin. Sister's husband and wife's brother = brother-in-law. Son's wife = daughter-in-law. Wife's or husband's father = father-in-law.

Draw the tree

Use ☐ for a male and ○ for a female. A horizontal line joins husband and wife; a vertical line goes down to children; people on the same horizontal row are of the same generation. Add each statement as one more link and never skip a step.



☐ male · ○ female · = married · vertical line = parent to child · same row = same generation

Father's brother = paternal uncle · Mother's brother = maternal uncle · Uncle's child = cousin

A family tree with the standard symbols

Reading 'pointing to' questions

'He is the son of my father's only son' means: my father's only son is my brother, so he is my brother's son = my nephew. Decode inside out: start from the innermost phrase.

Coded relations

Symbols stand for relations, for example A + B: A is the mother of B; A − B: A is the brother of B; A × B: A is the sister of B. Replace each symbol by its meaning in order and reduce step by step.

Formula and rule box

- Father's father = paternal grandfather
- Mother's brother = maternal uncle
- Father's sister = paternal aunt
- Brother's wife = sister-in-law
- Sister's husband = brother-in-law
- Uncle's or aunt's child = cousin
- 'My mother's only daughter' = myself or my sister (read carefully)

Step-by-step method

1. Convert each statement into a link on a tree.
2. Mark gender wherever it is known.
3. Work from the innermost phrase outwards.
4. Read the final relation directly from the tree.

Solved examples

Example 1. Pointing to a man, Ravi says 'He is the son of my mother's only brother.' How is the man related to Ravi?

Solution: Mother's brother = maternal uncle. His son = cousin. So the man is Ravi's cousin.

Answer: Cousin

Example 2. A is B's sister, C is B's mother and D is C's father. How is A related to D?

Solution: D is C's father, and C is the mother of A and B. So A is D's granddaughter.

Answer: Granddaughter

Example 3. Pointing to a photograph a woman says 'He is the son of my father's only son.' How is he related to her?

Solution: Father's only son = her brother. His son = her nephew.

Answer: Nephew

Example 4. If A + B means A is the mother of B, A − B means A is the brother of B, then P + Q − R means?

Solution: P is the mother of Q; Q is the brother of R. So P is the mother of R.

Answer: P is the mother of R

Shortcuts

- Start with the person who is mentioned most often.
- When gender is unknown, use a neutral word like 'child' or 'cousin' in your tree.
- 'Only' means exactly one, which fixes the relation.

Common mistakes

- Skipping the generation line in the tree.
- Assuming gender when the question does not give it.
- Mixing paternal and maternal sides.

Practice questions: Blood Relations

1. Father's brother's son is called? Basic
2. Mother's mother is called? Basic
3. A's mother is B's sister. How is B related to A? Basic
4. Pointing to a boy, Neha says 'He is the son of my mother's only daughter.' Relation? Basic
5. X is Y's brother, Z is Y's father and W is Z's father. How is X related to W? Medium
6. A is the brother of B, B is the sister of C, C is the father of D. How is A related to D? Medium
7. Pointing to a man a woman says 'His mother is the only daughter of my mother.' How is the woman related to the man? Medium
8. A is B's brother. C is A's mother. D is C's father. How is B related to D? Medium
9. P × Q + R means P is the sister of Q and Q is the mother of R. How is P related to R? Advanced
10. A is the father of B and C. C's son is D. How is A related to D? Advanced

Answers: see the answer key at the end of this guide.

04 REASONING ABILITY · IMPORTANT
Directions and Distances

Direction questions track a walker on a map. Plotting each move on a small grid turns every question into a picture you can measure.

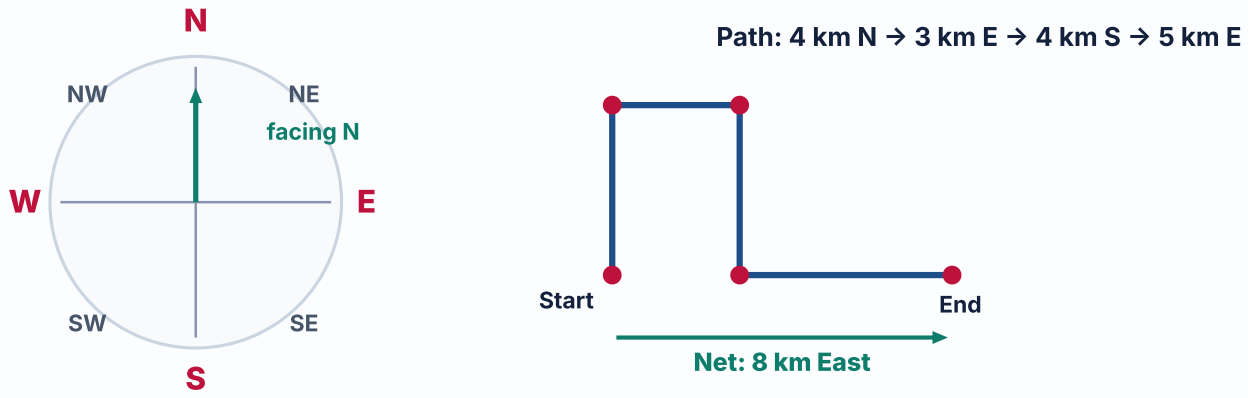
Start from zero

The compass

Four main directions are North, East, South and West; they are 90° apart going clockwise N → E → S → W. Between them are North-East, South-East, South-West and North-West at 45°. A turn to the right is a clockwise rotation and a turn to the left is anticlockwise.

Plot the path

Put the start at the origin. Draw each leg as an arrow on squared paper, using north as up and east as right. Net displacement = straight line from the start to the end.



Compass directions and the net displacement of a path

Net displacement and Pythagoras

If the final position is x km east and y km north, the straight-line distance = $\sqrt{(x^2 + y^2)}$. Triples like 3-4-5, 5-12-13, 8-15-17 and 6-8-10 appear again and again.

Shadow questions

In the morning the sun is in the east, so a shadow falls toward the west. At sunset the sun is in the west, so a shadow falls toward the east. At noon the shadow is shortest and points north (in the northern hemisphere).

Facing and turning

A right turn from North faces East, then South, then West. A left turn from North faces West, then South, then East. A 180° turn reverses the direction; 270° clockwise equals a 90° left turn.

Formula and rule box

- Right turn: N → E → S → W → N (clockwise)
- Left turn: N → W → S → E → N (anticlockwise)
- Distance = $\sqrt{(x^2 + y^2)}$
- Common triples: 3-4-5, 5-12-13, 8-15-17
- Morning shadow: West. Evening shadow: East
- Opposite directions cancel (north and south)

Step-by-step method

1. Draw a small compass and mark the start.
2. Draw each move; north is up, east is right.
3. Cancel opposite movements.
4. Use Pythagoras for the straight-line distance and name the final direction.

Solved examples

Example 1. A man walks 5 km north and then 12 km east. How far is he from the start?

Solution: Distance = $\sqrt{(5^2 + 12^2)} = \sqrt{169} = 13$ km (north-east side).

Answer: 13 km

Example 2. A man faces north, turns right, then right again and then left. Which direction does he now face?

Solution: North → right → East → right → South → left → East.

Answer: East

Example 3. Rohan walks 10 m west, turns right and walks 6 m, then turns right and walks 10 m. How far and in what direction is he from the start?

Solution: West 10, right (north) 6, right (east) 10. East and west cancel, leaving 6 m north.

Answer: 6 m north

Example 4. Rita walks 4 km north, 3 km east, 4 km south and 5 km east. Find the distance from the start.

Solution: North and south cancel. East = 3 + 5 = 8 km.

Answer: 8 km east

Shortcuts

- Opposite moves cancel before you apply Pythagoras.
- Keep a mini compass in the corner of your rough sheet.
- A 270° clockwise turn is the same as a 90° left turn.

Common mistakes

- Treating a left turn as a right turn.
- Forgetting the person's facing changes after each turn.
- Adding distances in perpendicular directions instead of using Pythagoras.

Practice questions: Directions and Distances

1. A walks 3 km south and then 4 km east. Straight-line distance? Basic
2. Facing north, a person turns left. New direction? Basic
3. Facing east, a person turns 90° right. New direction? Basic
4. A walks 8 km north, 6 km west and 8 km south. Distance from start? Basic
5. Facing south-west, a person turns 90° clockwise. New direction? Medium
6. Facing east, a person turns 180°. New direction? Medium
7. P goes 15 m north and then 20 m east. Distance from start? Medium
8. A walks 6 km east, 8 km north and 6 km west. Where is he from the start? Medium
9. At sunset, the shadow of a pole falls toward which direction? Advanced
10. A walks 4 km north, turns right for 3 km, right again for 4 km, then left for 5 km. Distance from start? Advanced

Answers: see the answer key at the end of this guide.

05 REASONING ABILITY · MUST-DO
Seating Arrangements

Seating puzzles give you a set of clues about who sits where. They are worth several questions in one go, so it pays to set them up cleanly.

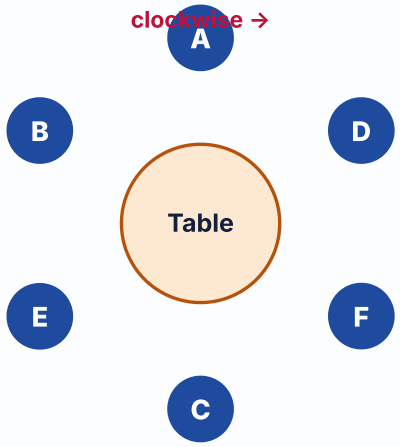
Start from zero

Linear arrangements

People sit in a row, usually facing north or south. If everyone faces north, the person's left is your left. If a row faces south, left and right swap relative to the page. Start with the clue that fixes a position (end, middle, 'second from left').

Circular arrangements

When people sit in a circle facing the centre, a person's **right is the anticlockwise neighbour and left is the clockwise neighbour**. If they face outside, it is the reverse. In an arrangement of 2n people, the opposite of position p is p + n.



Facing the centre:

- Right hand side = anticlockwise neighbour
- Left hand side = clockwise neighbour

Facing outside: it is the opposite.

Example order clockwise: A, D, F, C, E, B
Left of C (clockwise) = E · Right of C = F



Row facing north: left end → right end = Q P R S T

Left and right in circular and linear arrangements

Reading the clues

Direct clues ('A sits opposite B') are definite. Relative clues ('C is second to the right of A') need counting. Negative clues ('C is not next to A') rule options out. Use direct clues first, then relative ones, then negatives.

Table method

Draw empty seats numbered 1 to n. Write each fixed clue in the seat. When a clue allows two options, draw both and eliminate with later clues.

Counting persons

In a row of n people, the number between positions p and q is $|p - q| - 1$. Position from the right = n − (position from left) + 1.

Formula and rule box

- Opposite of seat p in a circle of 2n = p + n (mod 2n)
- Facing centre: right = anticlockwise, left = clockwise
- Facing outside: right = clockwise, left = anticlockwise
- Persons between positions p and q = $|p - q| - 1$
- Position from right = n − p + 1
- Number of arrangements around a circle = (n − 1)!

Step-by-step method

1. Draw the seats and number them.
2. Place the definite clues first.
3. Add relative clues by counting left and right correctly.
4. Use eliminations to fill the rest and answer.

Solved examples

Example 1. P, Q, R, S, T sit in a row facing north. R is in the middle, P is immediately left of R, T is at the right end and Q is not next to R. Who is between P and S?

Solution: Positions 1 to 5 from the left. R = 3, P = 2, T = 5. Remaining seats 1 and 4: Q cannot be next to R, so Q = 1 and S = 4. Order Q P R S T. Between P and S is R.

Answer: R

Example 2. Six people A–F sit around a table facing the centre. B is immediately right of A, C is opposite A, D is immediately left of A and E sits between C and B. Who is immediately left of C?

Solution: Label seats clockwise 1 to 6: A = 1. Right of A = anticlockwise = seat 6 = B. Left of A = seat 2 = D. C opposite = seat 4. E between C and B = seat 5, so F = 3. Left of C (clockwise) = seat 5 = E.

Answer: E

Example 3. In a row of 14 people, A is 6th from the left and 9th from the right. Check the total.

Solution: Total = 6 + 9 − 1 = 14.

Answer: 14

Example 4. Eight people sit equally spaced around a circle facing the centre. How many sit between a person and the person opposite on one side?

Solution: Opposite is 4 seats away, so 3 people between them.

Answer: 3

Shortcuts

- Solve a 'row' puzzle by writing the positions 1 to n and filling them.
- In a circle, label seats clockwise to avoid left–right confusion.
- Use definite clues (opposite, extreme end) before conditional ones.

Common mistakes

- Mixing up left and right when people face the centre.
- Starting with a negative clue.
- Not checking the final arrangement against every clue.

Practice questions: Seating Arrangements

1. 8 persons sit equally spaced in a circle. Who is opposite the 3rd person? Basic
2. In a row of 7 people P is exactly in the middle. His position from the left? Basic
3. Five persons A–E sit in a row facing north. C is at the extreme right, A is to the right of B, D is to the left of B and E is between A and C. Who is in the middle? Basic
4. Four friends P, Q, R, S face the centre of a square table. P is opposite R and Q is immediately right of P. Who is to the left of P? Basic
5. A is 6th from the left and 9th from the right in a row. Total persons? Medium
6. In a circle of 10 persons facing the centre, A is 3rd to the left of B. Then B is how many places to the right of A? Medium
7. P is 3rd from the left, Q is third to the right of P and R is second to the left of P. How many persons are between R and Q? Medium
8. 8 friends are equally spaced around a circle. How many sit between A and E who are opposite each other (one side)? Medium
9. In a row of 21 boys, Ramesh is 7th from the left and Suresh is 5th from the right. How many are between them? Advanced
10. Six people A–F face the centre. A is opposite D, B is immediately right of A, E is between D and B, C is not adjacent to A. Who is immediately right of D? Advanced

Answers: see the answer key at the end of this guide.

06 Syllogisms

REASONING ABILITY · MUST-DO

A syllogism gives two or three statements and asks which conclusions must be true. The safest method is to draw Venn diagrams.

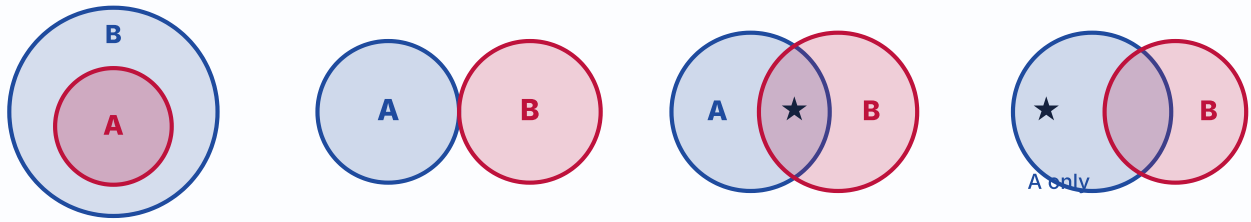
Start from zero

The four statement types

All A are B: A is fully inside B. **No A is B:** the circles are separate. **Some A are B:** the circles overlap. **Some A are not B:** at least part of A lies outside B. Each type has fixed conversions.

What converts

'No A is B' = 'No B is A'. 'Some A are B' = 'Some B are A'. 'All A are B' converts only to 'Some B are A'. 'Some A are not B' does not convert.



All A are B
A inside B

No A is B
separate circles

Some A are B
overlap (★ = common)

Some A are not B
★ in A outside B

Draw the Venn diagram of the statements. A conclusion is TRUE only if it holds in every possible diagram.

Four statement types as Venn diagrams

Venn diagram method

Draw one diagram that satisfies the statements. Then test the conclusion. A conclusion is **definitely true** only if it is true in every diagram you can draw. If it is true in one drawing and false in another, it does not follow.

Possibility and either-or

A conclusion that merely could be true is a 'possibility'. When two conclusions are a contradictory pair (for example 'Some A are B' and 'No A is B'), and neither is definite, then 'either I or II follows'.

Common results

All A are B and All B are C gives All A are C. Some A are B and All B are C gives Some A are C. All A are B and No B is C gives No A is C. Some A are B and No B is C gives Some A are not C. Two 'some' statements give no definite conclusion.

Formula and rule box

- All + All → All
- Some + All → Some
- All + No → No
- Some + No → Some not
- Some + Some → no conclusion
- No + No → no conclusion
- 'All A are B' → 'Some B are A'

Step-by-step method

1. Draw the Venn diagram of the first statement.
2. Add the second statement keeping the first true.
3. Test each conclusion in the diagram; try to break it.
4. Mark follows or does not follow, and use 'either or' for complementary pairs.

Solved examples

Example 1. Statements: All cats are dogs; All dogs are birds. Conclusions: I. All cats are birds. II. Some birds are cats.

Solution: Cats are inside dogs, which are inside birds. I is true. All cats are birds converts to Some birds are cats, so II is true.

Answer: Both I and II follow

Example 2. Statements: Some pens are books; All books are pencils. Conclusions: I. Some pens are pencils. II. All pens are pencils.

Solution: The pens that are books are pencils, so I is true. Not all pens are books, so II is not certain.

Answer: Only I follows

Example 3. Statements: No man is a bird; All birds are animals. Conclusions: I. No man is an animal. II. Some animals are not men.

Solution: Birds are animals and no bird is a man, so some animals are not men: II follows. Men could still be animals, so I does not follow.

Answer: Only II follows

Example 4. Statements: Some A are B; No B is C. Conclusion: Some A are not C.

Solution: The A that are B cannot be C, so Some A are not C.

Answer: Follows

Shortcuts

- Learn the six common results by heart.
- Prefer 'Some' over 'All' when in doubt.
- Test conclusions by trying to break them with another diagram.

Common mistakes

- Reversing 'All A are B' into 'All B are A'.
- Treating 'Some' as 'All'.
- Using outside knowledge instead of the statements.

Practice questions: Syllogisms

1. All roses are flowers; All flowers are plants. I. All roses are plants. II. Some plants are roses. Basic

2. Some doctors are teachers; All teachers are singers. I. Some doctors are singers. II. All doctors are singers. Basic

3. No pen is paper; Some papers are books. I. Some books are not pens. II. No book is a pen. Basic

4. All A are B; No B is C. I. No A is C. II. Some C are not A. Basic

5. Some A are B; Some B are C. I. Some A are C. II. Some C are A. Medium

6. All A are B; Some C are B. I. Some C are A. II. All C are B. Medium

7. Some A are B; All B are C; No C is D. I. Some A are C. II. Some A are not D. Medium

8. All A are B. I. Some B are A. II. No B is A. Medium

9. Some A are B. I. Some B are A. II. All A are B. Advanced

10. No A is B; No B is C. I. No A is C. II. Some A are C. Advanced

Answers: see the answer key at the end of this guide.

07 REASONING ABILITY · IMPORTANT

Clock and Calendar

Clock and calendar questions look different but both rest on one idea: relative movement. On a clock it is the speed of the two hands; on a calendar it is the number of leftover days after complete weeks.

Start from zero

The clock face

The full circle is 360° with 12 hours, so each hour mark is 30° apart and each minute mark is 6°. The minute hand moves 6° per minute and the hour hand moves 0.5° per minute, so the minute hand gains 5.5° per minute on the hour hand.

Angle between hands

Angle = | 30H − 5.5M | where H is the hour and M the minutes. If the result is more than 180°, take 360° minus it. At 3:30: |90 − 165| = 75°. At 4:20: |120 − 110| = 10°.



3:30 → angle 75°

Minute hand: 6° per minute

Hour hand: 30° per hour = 0.5° per minute

Angle = | 30H − 5.5M |

At 3:30 → |90 − 165| = 75°

If the angle is above 180°, subtract it from 360°.

Hands overlap 11 times in 12 hours and are at right angles 22 times in 24 h

Angle between clock hands

Overlaps and right angles

The hands overlap 11 times in 12 hours and 22 times in 24 hours. They are at right angles 22 times in 12 hours and 44 times in 24 hours. Between H and H+1 o'clock, the hands overlap at M = (60/11) × H minutes past H. At 4 o'clock this is 240/11 = 21 9/11 minutes.

Gaining and losing clocks

If a clock gains or loses a fixed time per hour, find how far it is off after the actual elapsed time. A clock that loses 10 minutes per hour shows 50 minutes for every actual 60 minutes.

Odd days

Count the days beyond complete weeks. An ordinary year has 1 odd day and a leap year has 2. 100 years have 5 odd days, 200 have 3, 300 have 1 and 400 have 0. Add odd days of complete years, complete months and the date, then take the remainder mod 7.

Day code and leap years

Sun = 0, Mon = 1, Tue = 2, Wed = 3, Thu = 4, Fri = 5, Sat = 6. A year divisible by 4 is a leap year, but a century year must be divisible by 400 (1900 is not a leap year, 2000 is).

Odd days of each month (days beyond complete weeks, ordinary year)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
3	0	3	2	3	2	3	3	2	3	2	3

Feb has 0 odd days in an ordinary year and 1 in a leap year. Ordinary year = 1 odd day, leap year = 2 odd days.

Odd days in centuries: 100 years = 5 · 200 years = 3 · 300 years = 1 · 400 years = 0

Day code: Sun 0 · Mon 1 · Tue 2 · Wed 3 · Thu 4 · Fri 5 · Sat 6

Odd-day table for the calendar

Formula and rule box

- Angle = $|30H - 5.5M|$
- Minute hand: 6° per minute
- Hour hand: 0.5° per minute
- Overlap between H and H+1 at M = 60H/11
- Hands overlap 11 times in 12 hours
- Ordinary year = 1 odd day, leap year = 2
- 100 years = 5, 200 = 3, 300 = 1, 400 = 0 odd days

Step-by-step method

1. Choose clock or calendar technique.
2. For clocks, use $|30H - 5.5M|$ and fix angles above 180°.
3. For calendars, add odd days of years, months and days.
4. Convert the total mod 7 to a day using the day code.

Solved examples

Example 1. Find the angle between the hands at 3:30.

Solution: $|30 \times 3 - 5.5 \times 30| = |90 - 165| = 75°$.

Answer: 75°

Example 2. Find the angle between the hands at 4:20.

Solution: $|120 - 5.5 \times 20| = |120 - 110| = 10°$.

Answer: 10°

Example 3. What day was 26 January 1950?

Solution: Years completed 1949: 1600 → 0, 300 → 1, 49 years → 49 + 12 leap = 61 → 5. Date 26 → 5. Total 0 + 1 + 5 + 5 = 11 → 4 = Thursday.

Answer: Thursday

Example 4. Today is Monday. What day will it be after 100 days?

Solution: 100 ÷ 7 leaves 2, so Monday + 2 = Wednesday.

Answer: Wednesday

Shortcuts

- The hour hand moves 0.5° per minute: at H:M the hour hand is at 30H + 0.5M degrees.
- In 12 hours the hands are at right angles 22 times, not 24.
- Repeat the same day every 7 days; reduce large day counts mod 7.

Common mistakes

- Forgetting to take 360° minus the angle when it exceeds 180°.
- Treating 1900 as a leap year.
- Counting odd days of the current year as complete.

Practice questions: Clock and Calendar

1. Angle between hands at 6:00? Basic
2. Angle between hands at 9:15? Basic
3. Angle between hands at 2:40? Basic
4. How many times do the hands overlap in 12 hours? Basic
5. How many degrees does the hour hand move in 20 minutes? Medium
6. Today is Monday. Day after 59 days? Medium
7. Is the year 2100 a leap year? Medium
8. How many odd days are in 200 years? Medium
9. If 5 February 2023 was a Sunday, what day was 5 March 2023? Advanced
10. A clock set right at 8 AM loses 10 minutes per hour. What time does it show at actual 2 PM? Advanced

Answers: see the answer key at the end of this guide.

08 REASONING ABILITY · IMPORTANT
Ranking and Ordering

Ranking questions place people in a line by position, height, age or marks. One formula and a clear picture solve most of them.

Start from zero

Position from both ends

Total persons = (position from left) + (position from right) − 1. If a person is 4th from the left in a row of 12, they are 9th from the right (12 − 4 + 1).

Persons between two people

If one person is at position p and another at position q (both from the same end), the number between them is |p − q| − 1. If positions are from opposite ends, convert one to the other end first.

Row of 12 students: Ravi is 4th from the left



grey = from left · red = from right

Total persons = (rank from left) + (rank from right) − 1 → 4 + 9 − 1 = 12

Position from left plus position from right minus one equals the total

Interchange of positions

When two persons exchange places, the total stays the same. A person's new position equals the other person's old position. Use that to find the total.

Comparison ordering

From statements like 'A is taller than B', 'B is taller than C', write a chain A > B > C. Merge several statements into one chain and read off the tallest, shortest or middle.

Ranks in class

Rank from top + rank from bottom − 1 = total students. If someone is 5 ranks below Neha (rank 8 from top), their rank from top is 13.

Formula and rule box

- Total = (from left) + (from right) − 1
- Position from right = Total − Position from left + 1
- Between two persons = |p − q| − 1
- After interchange: new position = other's old position
- Chain: A > B > C means A is tallest
- Rank from bottom = Total − rank from top + 1

Step-by-step method

1. Draw a row of boxes or a chain.
2. Place the known positions.
3. Use total = left + right − 1.
4. For interchanges use the new position to find the old one.

Solved examples

Example 1. In a row of 25 students, A is 10th from the left. What is his position from the right?

Solution: 25 − 10 + 1 = 16.

Answer: 16th

Example 2. In a row of 30, Rita is 12th from the left and Sita is 9th from the right. How many are between them?

Solution: Sita from left = 30 − 9 + 1 = 22. Between = 22 − 12 − 1 = 9.

Answer: 9

Example 3. A student is 5th from the top and 12th from the bottom. How many students in total?

Solution: 5 + 12 − 1 = 16.

Answer: 16

Example 4. Ram is 7th from the left and Shyam is 12th from the right. After interchanging, Ram is 15th from the left. How many people are in the row?

Solution: Shyam's old position from left = 15. Total = 15 + 12 − 1 = 26.

Answer: 26

Shortcuts

- Total = sum of both positions minus 1 is the master formula.
- Always convert both positions to the same end before counting between.
- In height chains, put the tallest on the left.

Common mistakes

- Forgetting the −1 in total.
- Counting the people themselves as 'between'.
- Mixing from-left and from-right positions.

Practice questions: Ranking and Ordering

1. A row has 40 persons. X is 15th from the left. Position from the right? Basic
2. Rank 8 from top and 14 from bottom. Total students? Basic
3. A is taller than B, C is taller than A, D is shorter than B. Who is the tallest? Basic
4. In a row of 35, A is 12th from the left and B is 10th from the right. How many are between them? Basic
5. Rohit is 17th from either end of a row. Total persons? Medium
6. Rank 10 from top and 25 from bottom. Total students? Medium

7. Priya is 10th from the left and Anu is 15th from the right. After interchange Priya is 20th from the left. Total? Medium

8. A is taller than B but shorter than C; D is taller than C. Who is second tallest? Medium

9. In 60 students, how many are between the 10th from the front and the 12th from the back? Advanced

10. In a class of 40, Neha is 8th from the top and Rahul is 5 ranks below her. Rahul's rank from the bottom? Advanced

Answers: see the answer key at the end of this guide.

09 REASONING ABILITY · IMPORTANT
Alphabet Test

Alphabet-test questions ask about positions, gaps and rearrangements of letters. Knowing the alphabet table by heart makes them almost mechanical.

Start from zero

The alphabet table

A to Z are positions 1 to 26. Counting from the right end, the position is 27 minus the position from the left. Opposite pairs (A–Z, B–Y, ...) sum to 27. The landmark letters E, J, O, T, Y are at 5, 10, 15, 20, 25.

Counting forward and backward

The nth letter to the right of a letter at position p is p + n; to the left is p − n. 'The 7th letter from the right end' is at position 27 − 7 = 20 (T).

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Top: letter · middle (grey): position from left · bottom (red): position from right = 27 − position

Opposite letters add to 27: A–Z, B–Y, C–X ... M–N. Gold = positions 5, 10, 15, 20, 25 (E J O T Y). Green = vowels.

Alphabet position table with the landmarks

Letters between

The number of letters between two letters at positions a and b is |a − b| − 1. The letter midway between two letters is at (a + b)/2 when the sum is even.

Rearranging words

For 'arrange the letters alphabetically' questions, write the letters in order and read the required position. Duplicate letters count as separate letters.

Meaningful words and pairs

A typical pattern asks you to form a meaningful word from given letters and name its first or last letter, or count pairs of letters having as many letters between them in the word as in the alphabet.

Formula and rule box

- Position from right = 27 − position from left
- nth letter right of position p = p + n
- Letters between two letters = |a − b| − 1
- Midway letter = (a + b)/2
- Landmarks: E5 J10 O15 T20 Y25
- Opposite letters sum to 27

Step-by-step method

1. Write positions of the letters involved.
2. Do the arithmetic on positions.

3. Convert back to letters.
4. For arrangement questions, sort first then count.

Solved examples

Example 1. Which letter is 5th to the right of the 10th letter from the left?

Solution: 10th = J. 10 + 5 = 15 = O.

Answer: O

Example 2. Find the opposite of M.

Solution: 27 − 13 = 14 = N.

Answer: N

Example 3. Which letter is exactly midway between C and K?

Solution: Positions 3 and 11, midway (3 + 11)/2 = 7 = G.

Answer: G

Example 4. Arrange the letters of BRAIN alphabetically. Which is the third letter?

Solution: A, B, I, N, R. The third letter is I.

Answer: I

Shortcuts

- Memorise EJOTY and count from the nearest landmark.
- From the right end use 27 − position.
- Write the word alphabetically before answering position questions.

Common mistakes

- Forgetting that the position from the right differs from the position from the left.
- Counting the end letters in 'between' questions.
- Skipping duplicate letters in arrangement.

Practice questions: Alphabet Test

1. Which letter is the 7th from the right end of the alphabet? Basic
2. Which letter is midway between F and P? Basic
3. Find the opposite letter of Q. Basic
4. Which letter is the 5th to the left of M? Basic
5. How many letters are between A and Z? Medium
6. Arrange the letters of STORAGE alphabetically. 4th letter? Medium
7. How many vowels are in COMPUTER? Medium
8. If the alphabet is reversed, which is the 6th letter from the left? Medium
9. Which letter is 3 places after the 5th letter from the right end? Advanced
10. Which letter is the 10th to the right of the 12th from the right end? Advanced

Answers: see the answer key at the end of this guide.

10 REASONING ABILITY · IMPORTANT

Symbol Operations and Inequalities

Here symbols are given new meanings. Success comes from translating each symbol first and calculating afterwards.

Start from zero

Redefined operators

A question may say a ★ b = 2a + b. Replace a and b with the given numbers and compute: 3 ★ 4 = 2 × 3 + 4 = 10. Always follow the given definition, not the usual meaning of the symbol.

Swapped signs

When + means ×, − means ÷ and so on, rewrite the whole expression with the real signs before applying BODMAS. For example with + as ×, − as ÷, × as − and ÷ as +, the expression 8 + 3 × 4 − 2 ÷ 5 becomes 8 × 3 − 4 ÷ 2 + 5 = 27.

Inequality chains: read left to right and combine only in the same direction

A > B = C < D

A > B = C → A > C ✓ (same direction) · B = C < D → B < D ✓

A and D: A is bigger than C, D is bigger than C → no definite relation ×

Redefined operators: replace the symbol by the new meaning first, then apply BODMAS.

If + means ×, − means ÷, × means −, ÷ means + → 8 + 3 × 4 − 2 ÷ 5 = 8×3 − 4÷2 + 5 = 27

Inequality chains and redefined operators

Coded inequalities

Symbols stand for relations: for example P # Q means P > Q, P $ Q means P < Q and P @ Q means P = Q. Replace each by the actual sign, then combine.

Combining inequalities

You can combine two relations only if the direction matches. A > B = C gives A > C. A > B < C gives no relation between A and C. A = B = C gives A = C. A ≥ B > C gives A > C.

Either-or

If neither of two conclusions about the same pair is definite but they complete each other (for example X > Y and X ≤ Y), then 'either I or II is true'.

Formula and rule box

- A > B > C → A > C
- A > B < C → no relation
- A = B ≥ C → A ≥ C
- A ≥ B > C → A > C
- Same direction chains combine; opposite directions break
- Replace symbols first, then use BODMAS

Step-by-step method

1. Rewrite the symbols with their real meaning.
2. Evaluate the expression using BODMAS.
3. For inequality chains, combine only consistent directions.
4. Test every conclusion against the chain.

Solved examples

Example 1. If $a \# b = a^2 - b$, find $5 \# 3$.

Solution: Replace a = 5, b = 3: 25 − 3 = 22.

Answer: 22

Example 2. If + means ×, − means ÷, × means − and ÷ means +, find 12 + 3 − 4 × 2 ÷ 5.

Solution: Rewrite: 12 × 3 ÷ 4 − 2 + 5 = 36 ÷ 4 − 2 + 5 = 9 − 2 + 5 = 12.

Answer: 12

Example 3. Given A > B = C < D, is A > C?

Solution: A > B and B = C, so A > C. Yes.

Answer: True

Example 4. Given R ≥ S > T = U, is R > U?

Solution: R ≥ S > T gives R > T and T = U, so R > U.

Answer: True

Shortcuts

- Write the rewritten expression on rough paper before calculating.
- A chain with a 'peak' or 'valley' breaks the relation between the ends.
- Check = and ≥ carefully: they can turn 'greater' into 'greater or equal'.

Common mistakes

- Calculating with the original signs.
- Combining relations with opposite directions.
- Ignoring 'equal to' in the chain.

Practice questions: Symbol Operations and Inequalities

1. If $a * b = 3a - 2b$, find $4 * 5$. Basic
2. If + means − and − means +, find 20 + 8 − 3. Basic
3. If × means ÷ and ÷ means ×, find 24 × 6 ÷ 2. Basic
4. A > B, B ≥ C. Relation between A and C? Basic
5. P = Q < R. Relation between P and R? Medium
6. X ≥ Y ≥ Z. Is X > Z definitely true? Medium
7. If $a \oplus b = (a + b)/2$, find $8 \oplus 12$. Medium
8. A < B > C. Is A < C definitely true? Medium

9. If $a \nabla b = a^2 - b^2$, find $7 \nabla 4$. Advanced

10. $2 @ 3 = 13$, $3 @ 4 = 25$, $4 @ 5 = 41$. Find $5 @ 6$. Advanced

Answers: see the answer key at the end of this guide.

11 REASONING ABILITY · GOOD TO KNOW
Analogies and Odd One Out

These two topics test whether you can see the link between items. They take very little time once you know the common relation types.

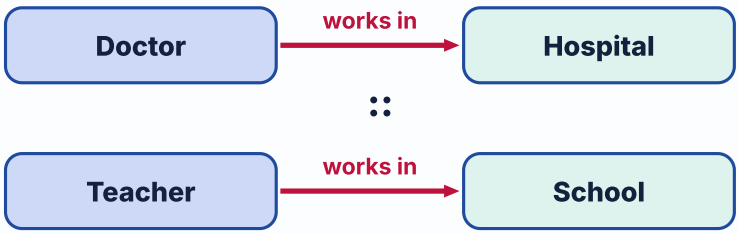
Start from zero

Analogy: A : B :: C : D

The relation between A and B must be the same as between C and D. First put the relation into a sentence ('A is the place where B works') and test it on the second pair.

Types of relation

Worker and workplace (Doctor : Hospital), tool and use (Pen : Write), cause and effect (Fire : Smoke), part and whole (Petal : Flower), synonym and antonym (Hot : Cold), category and example (Fruit : Mango). For numbers: squares (5 : 25), cubes (3 : 27), multiples. For letters: shifts and opposites.



Common analogy types

- Worker : workplace · Tool : use
- Cause : effect · Part : whole
- Synonym · Antonym
- Number: square, cube · Letter: shift

The relation in the first pair must match the second pair

Odd one out

Find what three of the items share and pick the one that does not belong. Look for type (living or non-living), shape (2D or 3D), maths (prime, square, cube, odd or even) or spelling and word-length patterns.

Number analogies

Look for squares, cubes, multiplication and addition links. Check whether the same rule works for the second pair: 3 : 27 gives n^3, so 4 : 64.

Formula and rule box

- Put the relation into one sentence
- Squares: n → n^2; Cubes: n → n^3
- Letter analogy: check the shift or opposite
- Odd one out: three share a property, one does not
- Typical categories: living / non-living, 2D / 3D, prime / composite

Step-by-step method

1. Read the first pair and state the relation.
2. Apply the same relation to the second pair.
3. For odd one out, list a common property for each group of three.
4. Eliminate options that break the rule.

Solved examples

Example 1. Doctor : Hospital :: Teacher : ?

Solution: Doctor works in a hospital, so a teacher works in a school.

Answer: School

Example 2. 3 : 27 :: 4 : ?

Solution: $3^3 = 27$, so $4^3 = 64$.

Answer: 64

Example 3. CAT : DBU :: DOG : ?

Solution: Each letter is +1: D→E, O→P, G→H.

Answer: EPH

Example 4. Find the odd one out: 8, 27, 64, 100, 125.

Solution: 8, 27, 64 and 125 are perfect cubes. 100 is not.

Answer: 100

Shortcuts

- Convert the relation into a sentence and test it.
- For number analogies test square and cube first.
- In odd one out, look first for a mathematical property.

Common mistakes

- Choosing a pair that fits loosely instead of the exact relation.
- Ignoring the order of the pair (Cause : effect vs Effect : cause).
- Over-thinking simple categories.

Practice questions: Analogies and Odd One Out

1. Bird : Fly :: Fish : ? Basic
2. Pen : Write :: Knife : ? Basic
3. Hot : Cold :: Day : ? Basic
4. 5 : 125 :: 4 : ? Basic
5. AZ : BY :: CX : ? Medium
6. Find the odd one out: 3, 5, 7, 9, 11. Medium
7. Find the odd one out: Square, Triangle, Circle, Cube. Medium
8. Find the odd one out: Lion, Tiger, Leopard, Cow. Medium
9. 6 : 216 :: 7 : ? Advanced
10. Find the odd one out: 121, 144, 169, 200, 225. Advanced

Answers: see the answer key at the end of this guide.

Cheat sheet

Every key rule and formula of the section on a few pages. Revise it daily in the last two weeks.

1. Number and Alphabet Series

- Arithmetic: next = last + d
- Geometric: next = last × r
- Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100
- Cubes: 1, 8, 27, 64, 125, 216
- Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
- Fibonacci: each term = sum of previous two
- Letter position: A=1 ... Z=26; opposite = 27 − position

2. Coding and Decoding

- Opposite letter = 27 − position
- Forward shift: new position = old + k (wrap after 26)
- Alternating shift: +k, −k, +k, ...
- Position value: A=1, B=2, ... Z=26
- Common-word method for sentence codes
- Check the first two letters to detect the rule

3. Blood Relations

- Father's father = paternal grandfather
- Mother's brother = maternal uncle
- Father's sister = paternal aunt
- Brother's wife = sister-in-law
- Sister's husband = brother-in-law
- Uncle's or aunt's child = cousin
- 'My mother's only daughter' = myself or my sister (read carefully)

4. Directions and Distances

- Right turn: N → E → S → W → N (clockwise)
- Left turn: N → W → S → E → N (anticlockwise)
- Distance = $\sqrt{(x^2 + y^2)}$
- Common triples: 3-4-5, 5-12-13, 8-15-17
- Morning shadow: West. Evening shadow: East
- Opposite directions cancel (north and south)

5. Seating Arrangements

- Opposite of seat p in a circle of 2n = p + n (mod 2n)
- Facing centre: right = anticlockwise, left = clockwise
- Facing outside: right = clockwise, left = anticlockwise
- Persons between positions p and q = $|p - q| - 1$
- Position from right = n − p + 1
- Number of arrangements around a circle = (n − 1)!

6. Syllogisms

- All + All → All
- Some + All → Some
- All + No → No
- Some + No → Some not
- Some + Some → no conclusion
- No + No → no conclusion
- 'All A are B' → 'Some B are A'

7. Clock and Calendar

- Angle = $|30H - 5.5M|$
- Minute hand: 6° per minute
- Hour hand: 0.5° per minute
- Overlap between H and H+1 at M = 60H/11
- Hands overlap 11 times in 12 hours
- Ordinary year = 1 odd day, leap year = 2
- 100 years = 5, 200 = 3, 300 = 1, 400 = 0 odd days

8. Ranking and Ordering

- Total = (from left) + (from right) − 1
- Position from right = Total − Position from left + 1
- Between two persons = $|p - q| - 1$
- After interchange: new position = other's old position
- Chain: A > B > C means A is tallest
- Rank from bottom = Total − rank from top + 1

9. Alphabet Test

- Position from right = 27 − position from left
- nth letter right of position p = p + n
- Letters between two letters = $|a - b| - 1$
- Midway letter = (a + b)/2
- Landmarks: E5 J10 O15 T20 Y25
- Opposite letters sum to 27

10. Symbol Operations and Inequalities

- A > B > C → A > C
- A > B < C → no relation
- A = B ≥ C → A ≥ C
- A ≥ B > C → A > C
- Same direction chains combine; opposite directions break
- Replace symbols first, then use BODMAS

11. Analogies and Odd One Out

- Put the relation into one sentence
- Squares: n → n^2; Cubes: n → n^3
- Letter analogy: check the shift or opposite
- Odd one out: three share a property, one does not
- Typical categories: living / non-living, 2D / 3D, prime / composite

Mini test: 20 questions in 25 minutes

Take this without looking at the guide. Target: 16 or more correct. No negative marking.

1. 6, 11, 21, 36, 56, ?
2. If HELP = IFMQ, code SAVE in the same way.
3. Pointing to a man, Ria says 'He is the brother of my father's wife.' How is he related to Ria?
4. A walks 12 m north and then 5 m east. Distance from the start?
5. In a row of 40, A is 14th from the left. Position from the right?
6. All A are B; All B are C. Does 'All A are C' follow?
7. Angle between the hands at 5:00?
8. 1 January 2024 was a Monday. What day was 1 January 2025?
9. Z, X, V, T, ?
10. Arrange the letters of COMPUTER alphabetically. The 7th letter?
11. If 3 @ 4 = 25 and 5 @ 12 = 169, find 8 @ 15.
12. Doctor : Patient :: Teacher : ?
13. Find the odd one out: 16, 25, 36, 48, 49.
14. 8 persons sit equally spaced in a circle. Who is opposite the 2nd person?
15. A student is 12th from the top among 30. Rank from the bottom?
16. If + means ×, × means −, − means ÷, ÷ means +, find 8 + 3 × 4 − 2 ÷ 5.
17. Reverse-alphabet code of BAD?
18. Today is Wednesday. What day after 45 days?
19. Facing east, a person turns right and then right again. Direction now?
20. Some A are B; No B is C. Does 'Some A are not C' follow?

Answer key

1. Number and Alphabet Series

1. 21
2. 48
3. 125
4. 37
5. 24
6. 17
7. O
8. 13
9. 252
10. 10

2. Coding and Decoding

1. MBNQ
2. WLT
3. 7
4. FQI
5. EHQD
6. 3 8 1 9 18
7. 57
8. NNPM
9. DTCKP
10. 214673

3. Blood Relations

1. Cousin
2. Grandmother (maternal)
3. Maternal aunt
4. Her son
5. Grandson
6. Uncle
7. Mother
8. Grandchild
9. Aunt
10. Grandfather

4. Directions and Distances

1. 5 km
2. West
3. South
4. 6 km west
5. North-west
6. West
7. 25 m
8. 8 km north
9. East
10. 8 km east

5. Seating Arrangements

1. 7th person
2. 4th
3. A
4. S
5. 14
6. 3rd
7. 4
8. 3
9. 9
10. C

6. Syllogisms

1. Both follow
2. Only I follows
3. Only I follows
4. Both follow
5. Neither follows
6. Neither follows
7. Both follow
8. Only I follows
9. Only I follows
10. Either I or II follows

7. Clock and Calendar

1. 180°
2. 172.5°
3. 160°
4. 11

5. 10°
6. Thursday
7. No
8. 3
9. Sunday
10. 1 PM

8. Ranking and Ordering

1. 26th
2. 21
3. C
4. 13
5. 33
6. 34
7. 34
8. C
9. 38
10. 28th

9. Alphabet Test

1. T
2. K
3. J
4. H
5. 24
6. O
7. 3
8. U
9. Y
10. Y

10. Symbol Operations and Inequalities

1. 2
2. 15
3. 8
4. A > C
5. P < R
6. No (only X ≥ Z)
7. 10
8. No (cannot be determined)
9. 33
10. 61

11. Analogies and Odd One Out

1. Swim
2. Cut
3. Night
4. 64
5. DW
6. 9
7. Cube
8. Cow
9. 343
10. 200

Mini test answers

1. 81
2. TBWF
3. Maternal uncle
4. 13 m
5. 27th
6. Yes
7. 150°
8. Wednesday
9. R
10. T
11. 289
12. Student
13. 48
14. 6th
15. 19th
16. 27
17. YZW
18. Saturday
19. West
20. Yes